

Exercises Chapter 4

1. Consider the nonparametric regression model

$$Y_i = f(X_i) + \epsilon_i, \quad i = 1, \dots, n,$$

where $X_i = i/n$, the error terms ϵ_i are i.i.d. with $E(\epsilon_i) = 0$ and $Var(\epsilon_i) = \sigma^2 > 0$, and f is an unknown smooth function. Suppose we estimate f using the KNN-regression estimator

$$\hat{f}_K(x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} Y_i.$$

at a fixed point $x_0 \in (0, 1)$, where $\mathcal{N}_0$ is the set of indices corresponding to the K nearest neighbors of x_0 among the design points $X_1, \dots, X_n$.

- (a) Compute the bias $Bias(\hat{f}_K(x_0)) = E(\hat{f}_K(x_0)) - f(x_0)$ of the KNN-regression estimator $\hat{f}_K(x_0)$.

- (b) Compute the variance $Var(\hat{f}_K(x_0))$ of the KNN-regression estimator $\hat{f}_K(x_0)$.

- (c) Now, let $f(x) = 5 + 3 \sin(2\pi x)$, $\sigma^2 = 1$, and $n = 100$.

Compute the bias of the KNN-regression estimator $\hat{f}_K(x_0)$ at $x_0 = 0.25$ for $K = 5$ and $K = 20$.

- (d) Consider the same setup as in the previous question.

Compute the variance of the KNN-regression estimator $\hat{f}_K(x_0)$ at $x_0 = 0.25$ for $K = 5$ and $K = 20$.

- (e) Consider the same setup as in the previous question.

Compute the local mean squared estimation error of the KNN-regression estimator $\hat{f}_K(x_0)$ at $x_0 = 0.25$ for $K = 5$ and $K = 20$.

- (f) Consider the same setup as in the previous question.

Compute the local mean squared prediction error of the KNN-regression estimator $\hat{f}_K(x_0)$ at $x_0 = 0.25$ for $K = 5$ and $K = 20$.

2. **Curse of Dimensionality:** The accuracy of nonparametric methods typically deteriorates as the dimension $p \geq 1$ of the predictor variables $X_i = (X_{i1}, \dots, X_{ip})^\top$ increases. This phenomenon is known as the "curse of dimensionality". The reason for this problem is the "emptiness" of high-dimensional spaces. To illustrate this, consider the case of a p -dimensional predictor variable $X_i = (X_{i1}, \dots, X_{ip})^\top$, with standard normal components $X_{ij} \sim \mathcal{N}(0, 1)$ for all $j = 1, \dots, p$ all independent from each other; i.e. X_{ij} and X_{ik} are independent for all $j \neq k$.

- (a) Consider the case of $p = 1$: Compute the probability that X_i lies in the interval $[-1, 1]$.

- (b) Consider the case of $p = 2$: Compute the probability that X_i lies in the square $[-1, 1]^2$.

- (c) Consider the general case of $p \geq 1$: Compute the probability that X_i lies in $[-1, 1]^p$.

- (d) Consider a sample size of $n = 100$. How many data points do you expect to see in $[-1, 1]^p$ for $p \in \{1, 5, 15\}$?